

Midterm 2 Review

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Contents

Course Syllabus and reading assignments:

Part I: Review of modeling and simulation (Chapters 1-3, 5)

- a. Introduction
- b. Modeling in the Laplace domain
- c. Transfer functions and block diagrams

Part II: Feedback control analysis (Chapter 4, 6-8)

- a. Time response analysis and specifications
- b. Stability analysis
- c. Steady-state errors
- d. Root-locus techniques

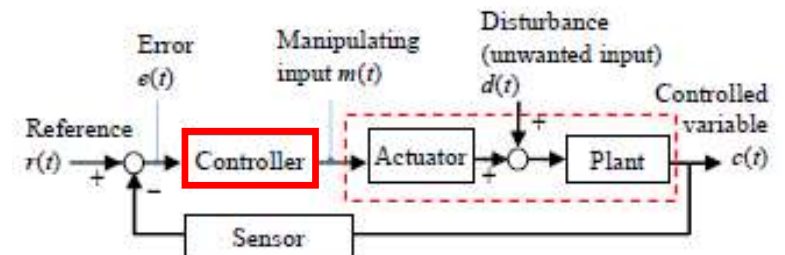
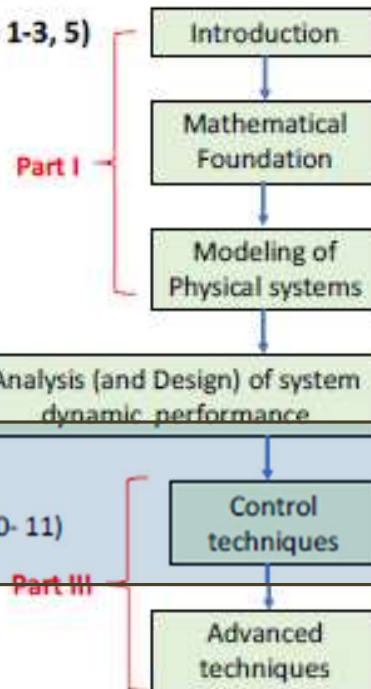
Mid-term 1

Part III: Feedback control design

- a. Root-locus design (Chapter 9)
- b. Frequency domain analysis and design (Chapter 10- 11)
- c. State-space design (Chapter 12)

Mid-term 2

- d. Digital control Systems (Chapter 13)

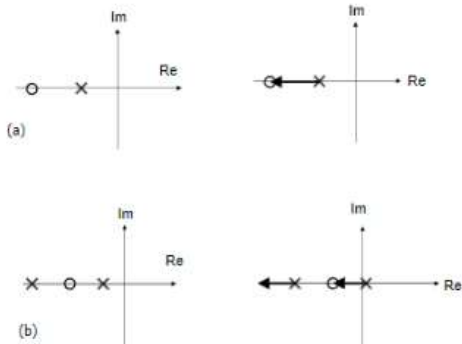


SISO Automatic (Closed-loop) Control System

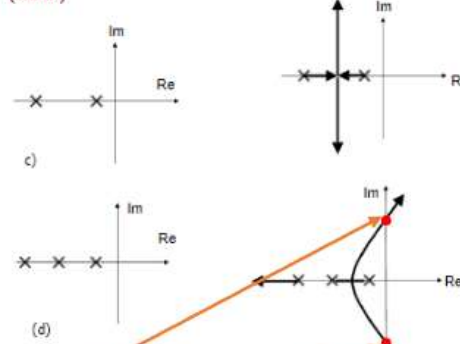
Control objective: $c(t) \rightarrow r(t)$ regardless of $d(t)$.

Root Locus Design

Exercise A: RL Quick Sketches without detailed calculations.



Exercise A: RL Quick Sketches without detailed calculations (Cont.)

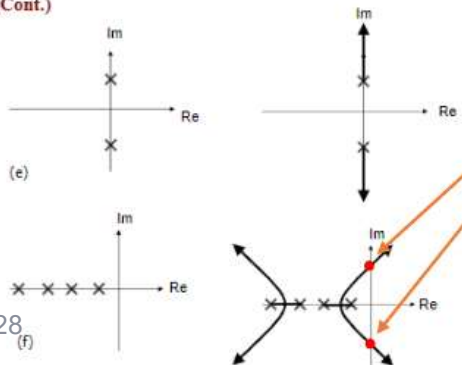


Points on RL satisfy two conditions:

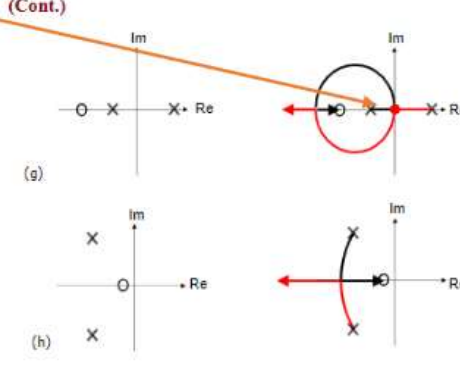
$$\text{Mag. } K \left| \frac{N(z)}{D(z)} \right| = 1 \Rightarrow K = \left| \frac{D(z)}{N(z)} \right|$$

$$\text{Phase } \angle \left[\frac{N(z)}{D(z)} \right] = \begin{cases} \pm 180(2I+1) & K \geq 0 \\ \pm 180(2I) & K < 0 \end{cases} \text{ where } i = 0, 1, 2, \dots$$

Exercise A: RL Quick Sketches without detailed calculations (Cont.)



Exercise A: RL Quick Sketches without detailed calculations (Cont.)

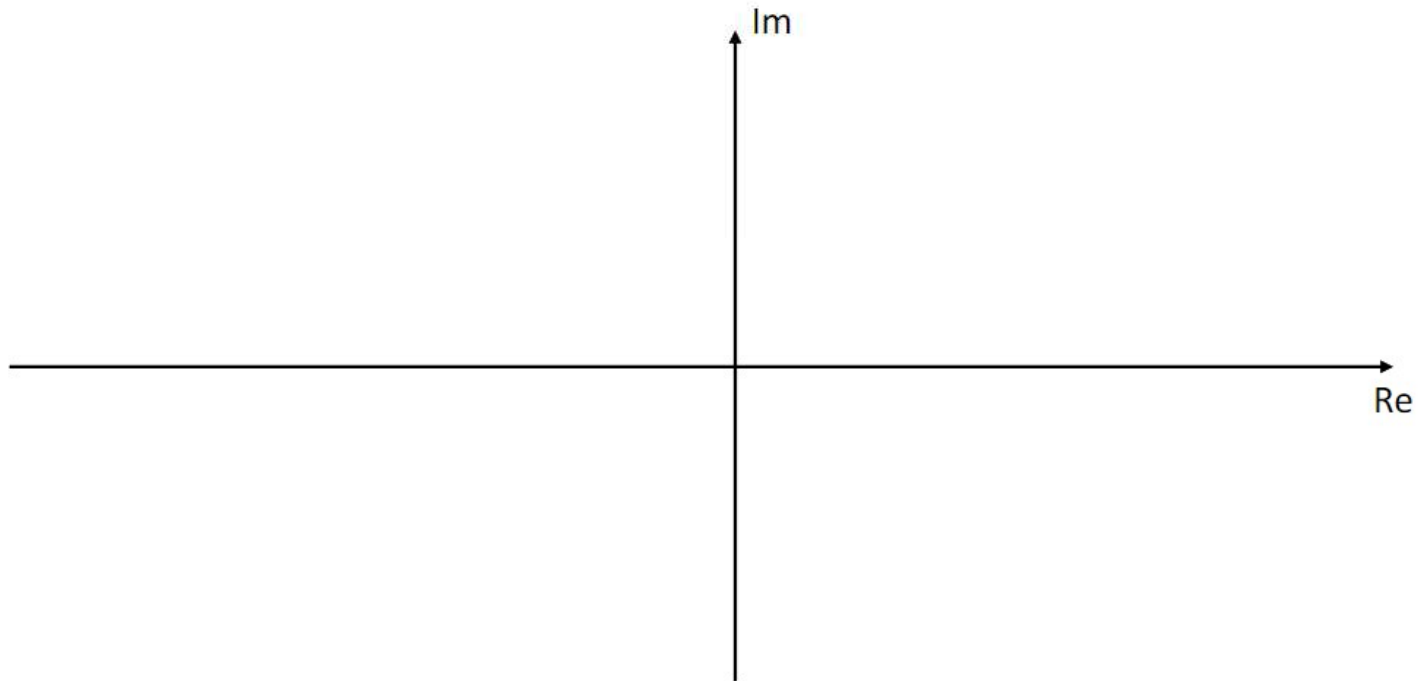


Critical Point

$$\sum \frac{1}{\sigma_b - p_i} = \sum \frac{1}{\sigma_b - z_i}$$

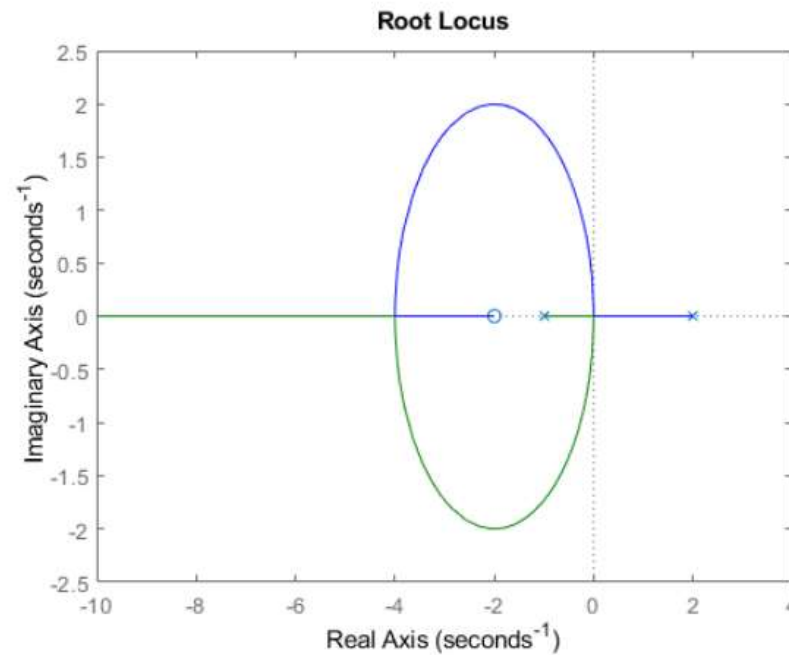
Example

$$G(s) = \frac{s+2}{(s-2)(s+1)}, \sigma_b = 4$$



Example

$$G(s) = \frac{s+2}{(s-2)(s+1)}, \sigma_b = 4$$



Bode Diagram

2 plots on logarithmic scales

Magnitude $20 \log_{10} |G(j\omega)|$ dB

Phase $\angle G(j\omega)$ degrees

Standard Forms

1st order lead (+), PD controller (High-pass filter): $Ts + 1$
 1st order lag (-), Mech damper (Low-pass filter): $1 / (Ts + 1)$ $G(s) = (Ts + 1)^{\pm 1}$

$$T\omega_c = 1$$

2nd order lead (+) and lag (-) :
 Example: mass-spring-damper system. $G(s) = \left(\frac{s^2}{\omega_n^2} + 2\zeta \frac{s}{\omega_n} + 1 \right)^{\pm 1}$

Routh Criterion Vs. Root Locus – CL poles' place in real-imaginary plane
 Bode Plot - IO magnitude ratio(gain) & phase shift
 Nyquist – IO magnitude ratio(gain) & phase shift
 in real-imaginary plane

Close loop

Open loop

“Critical gain from each criterion is the same”

Bode Diagram

Bode Asymptotic Plots (standard form)

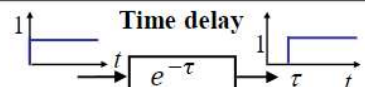
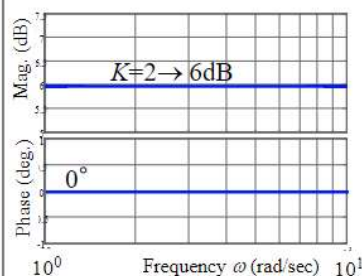
Constant Gain

$$G(s) = K \rightarrow G(j\omega) = K$$

Magnitude (dB):

$$20 \log |G(j\omega)| = 20 \log |K| \text{ dB}$$

Phase: $\angle G(j\omega) = \angle K = 0^\circ$



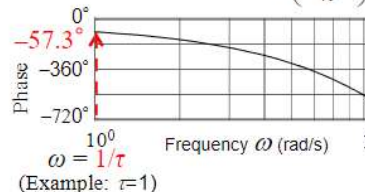
$$G(s) = e^{-\tau s} \Rightarrow G(j\omega) = e^{-\tau j\omega}$$

Magnitude (dB):

$$20 \log |G(j\omega)| = 0 \text{ dB}$$

Phase:

$$\angle G(j\omega) = \angle e^{-\tau j\omega} = -\tau\omega \left(\frac{180^\circ}{\pi} \right)$$



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Integrator

$$G(s) = \frac{1}{s} \rightarrow G(j\omega) = \frac{1}{j\omega}$$

Magnitude (dB):

$$20 \log \left| \frac{1}{j\omega} \right| = -20 \log \omega \text{ dB}$$

Phase: $\angle \left(\frac{1}{j\omega} \right) = -90^\circ$



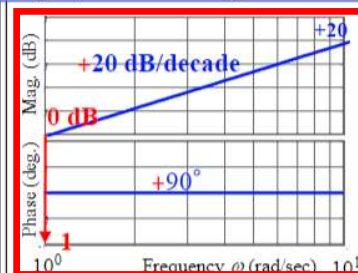
Derivative

$$G(s) = s \rightarrow G(j\omega) = j\omega$$

Magnitude (dB):

$$20 \log |j\omega| = +20 \log \omega \text{ dB}$$

Phase: $\angle j\omega = +90^\circ$



$$G(s) = s^{\pm 1} \quad \text{Mag: } \pm 20 \log \omega \text{ dB} \\ \rightarrow G(j\omega) = (j\omega)^{\pm 1} \quad \text{Phase: } \angle j\omega = \pm 90^\circ$$

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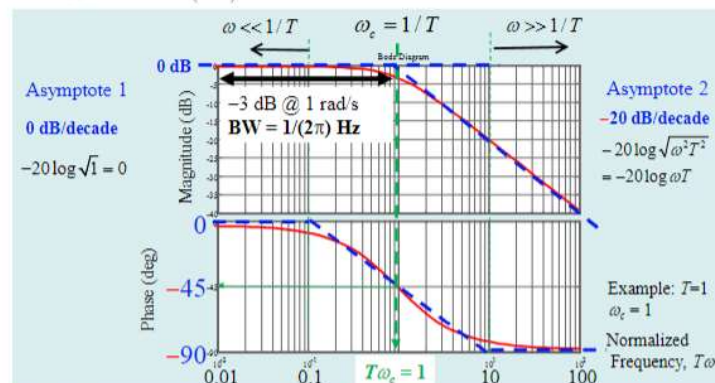
1st order System

$$G(s) = \frac{1}{Ts + 1} \quad G(j\omega) = \frac{1}{Tj\omega + 1}$$

Mag. (dB): $-20 \log \sqrt{(T\omega)^2 + 1}$

Phase ($^\circ$): $-\tan^{-1}(\omega T)$

	Mag. (dB)	Phase ($^\circ$)
$\omega \ll 1/T$	1 or 0 dB	0°
$\omega = 1/T$	-3 dB	-45°
$\omega \gg 1/T$	$-20 \log(\omega T)$	-90°



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Bode Asymptotic Plots: 2nd order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{1}{\frac{s^2}{\omega_n^2} + 2\zeta \frac{s}{\omega_n} + 1}$$

$$(j\omega)^2 = -\omega^2$$

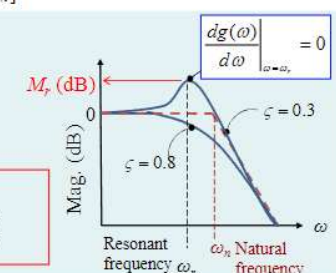
$$G(j\omega) = \frac{1}{[1 - (\omega/\omega_n)^2] + j(2\zeta\omega/\omega_n)} \quad g(\omega) = [1 - (\omega/\omega_n)^2]^2 + (2\zeta\omega/\omega_n)^2$$

Mag. (dB): $-20 \log \sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\zeta\omega/\omega_n]^2}$

Phase ($^\circ$): $-\tan^{-1} \frac{2\zeta\omega/\omega_n}{1 - (\omega/\omega_n)^2}$

$$\left. \frac{dg(\omega)}{d\omega} \right|_{\omega=\omega_r} = 0 \rightarrow \omega_r = \omega_n \sqrt{1 - 2\zeta^2}$$

$$M_r = 20 \log |G(j\omega)| \\ M_r = -20 \log (2\zeta \sqrt{1 - 2\zeta^2}) \begin{cases} > 0 & \zeta > 0.7 \\ < 0 & \zeta < 0.7 \end{cases}$$

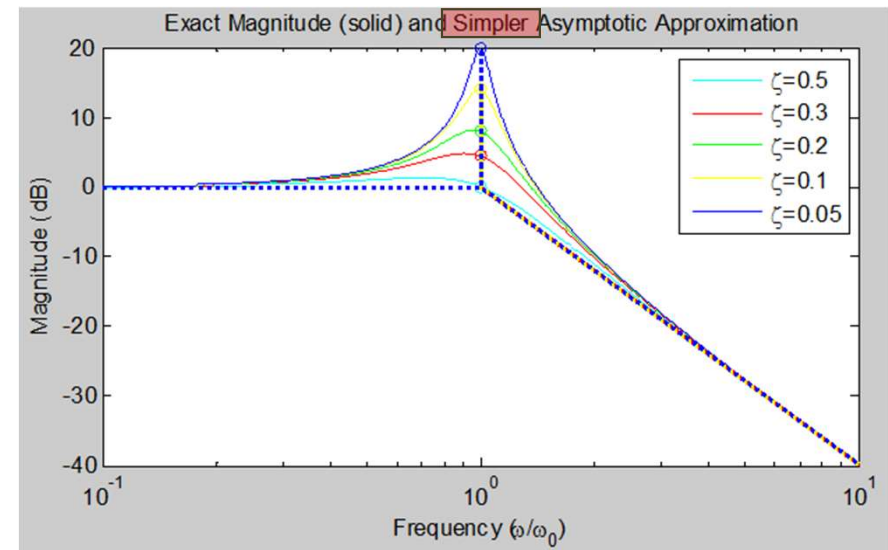
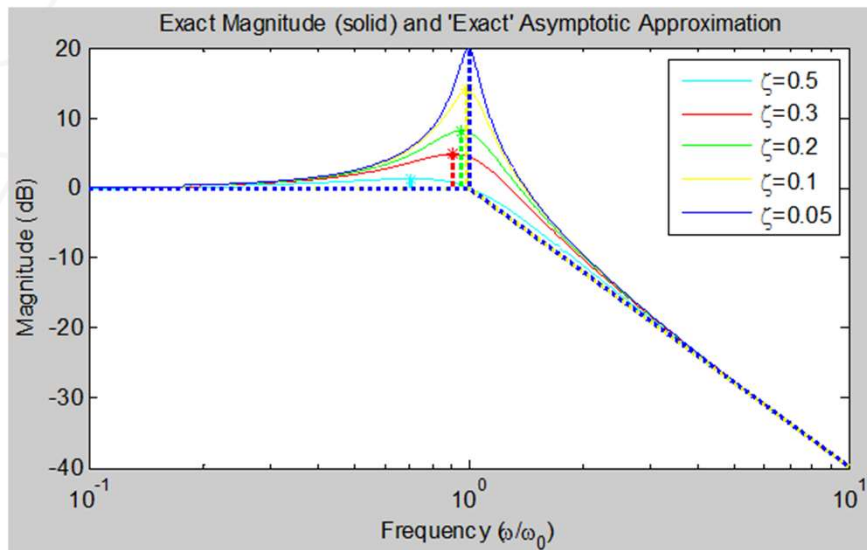


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Position of Curve:

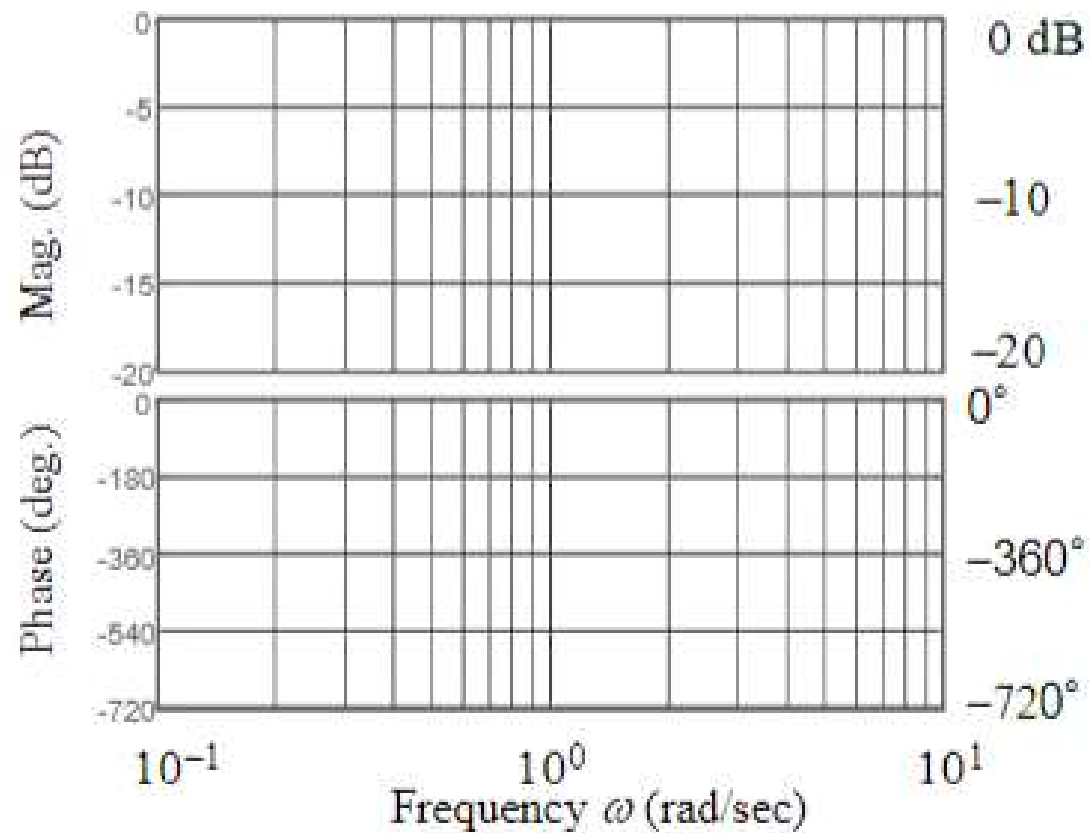
$$\frac{1}{10\omega_c} \text{ & } 10\omega_c$$

Bode Diagram

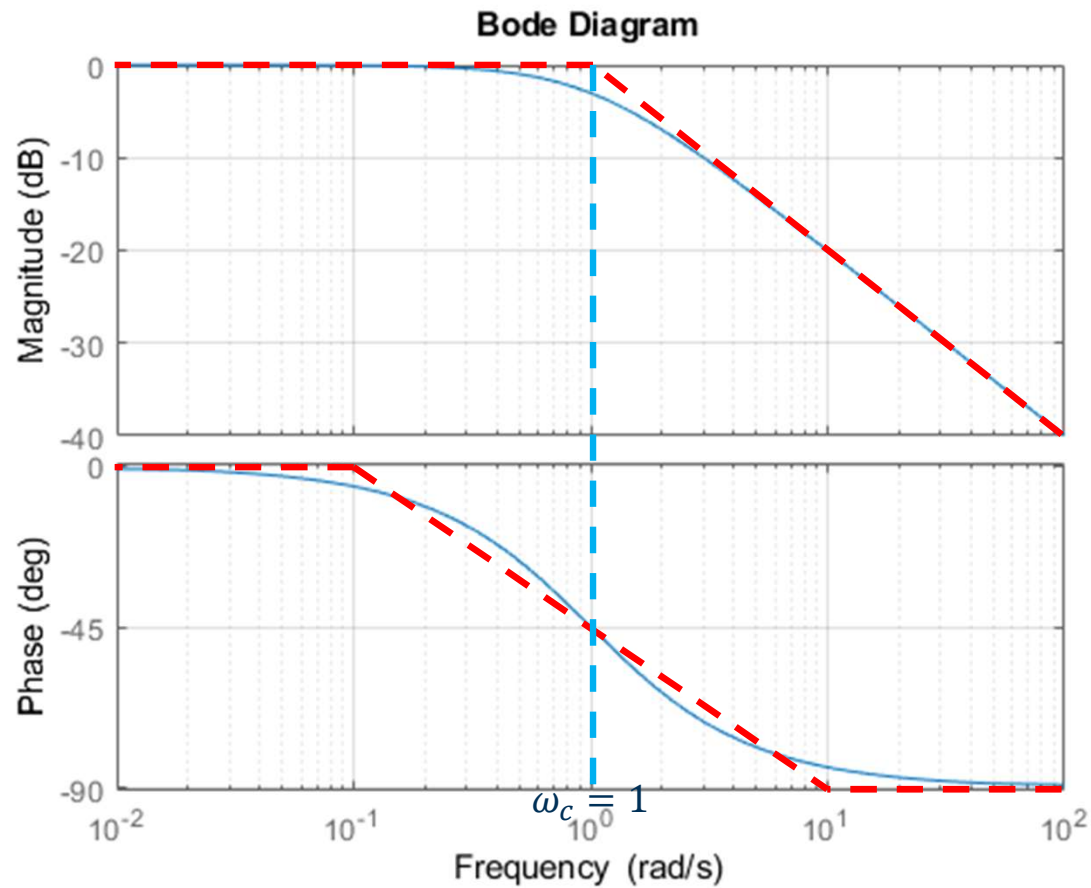


Using simpler approximation is okay

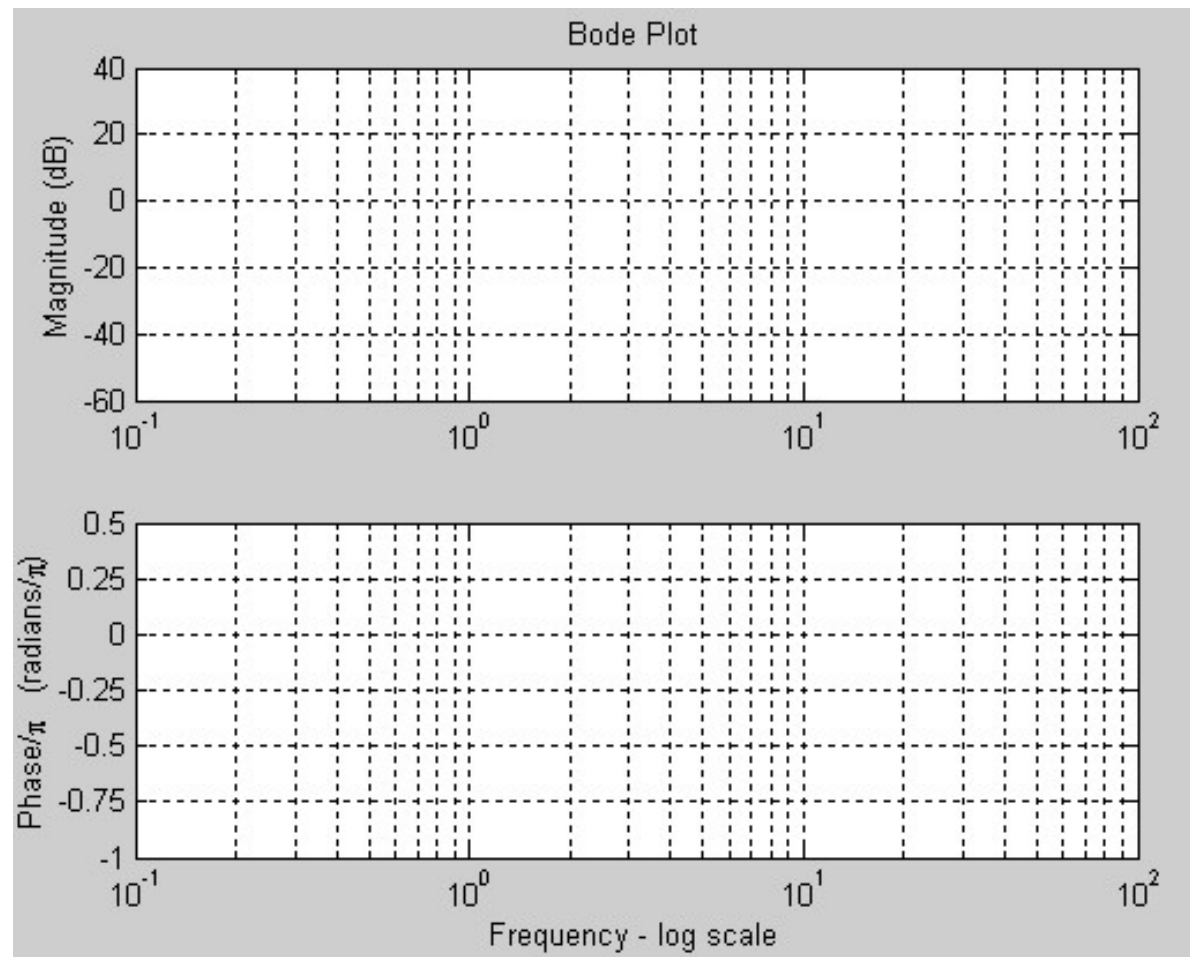
Example - $G(s) = \frac{1}{s+1}$



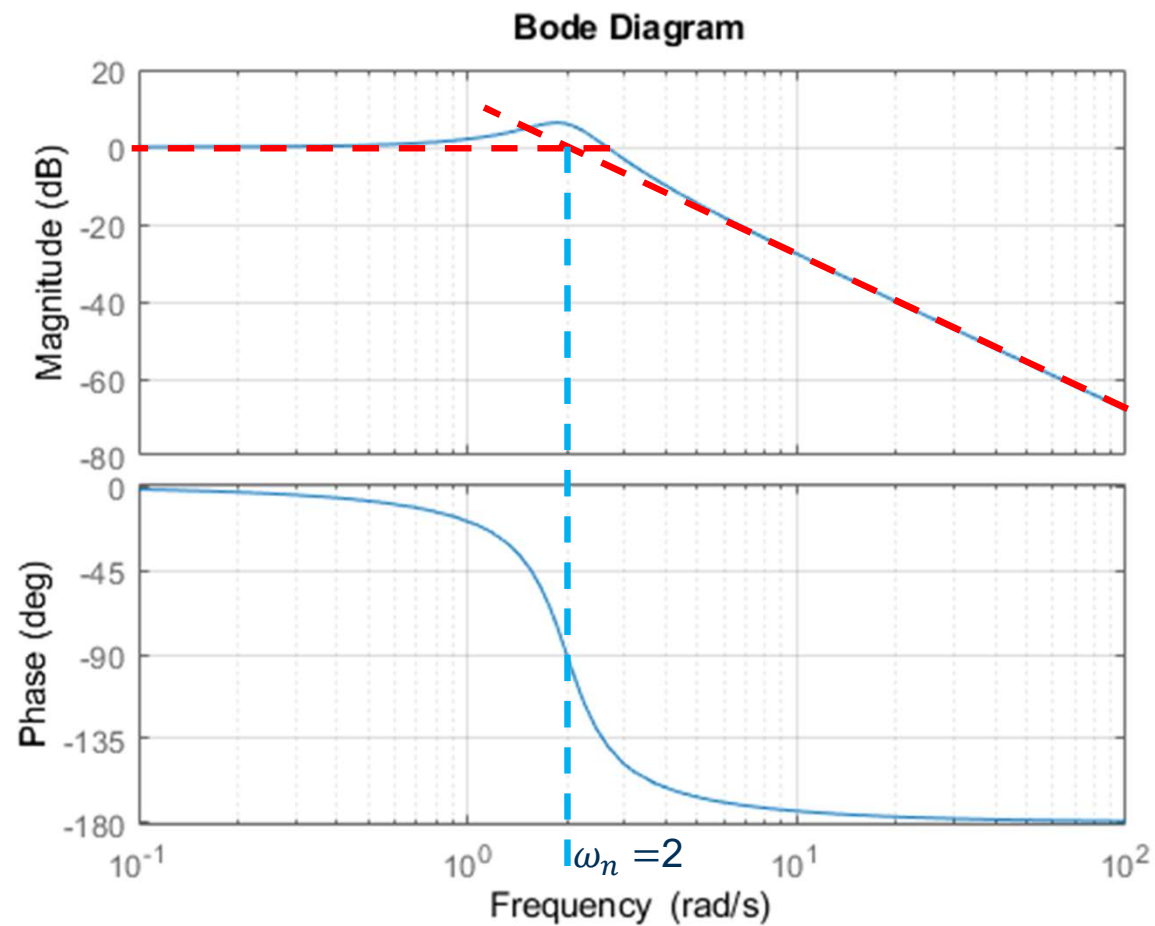
Example - $G(s) = \frac{1}{s+1}$



Example - $G(s) = \frac{4}{s^2 + s + 4}$



Example - $G(s) = \frac{4}{s^2 + s + 4}$

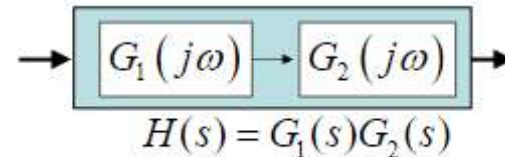


Bode Diagram - Superposition

<https://github.com/echeever/BodePlotGui>

Properties of Bode plots (cont.)

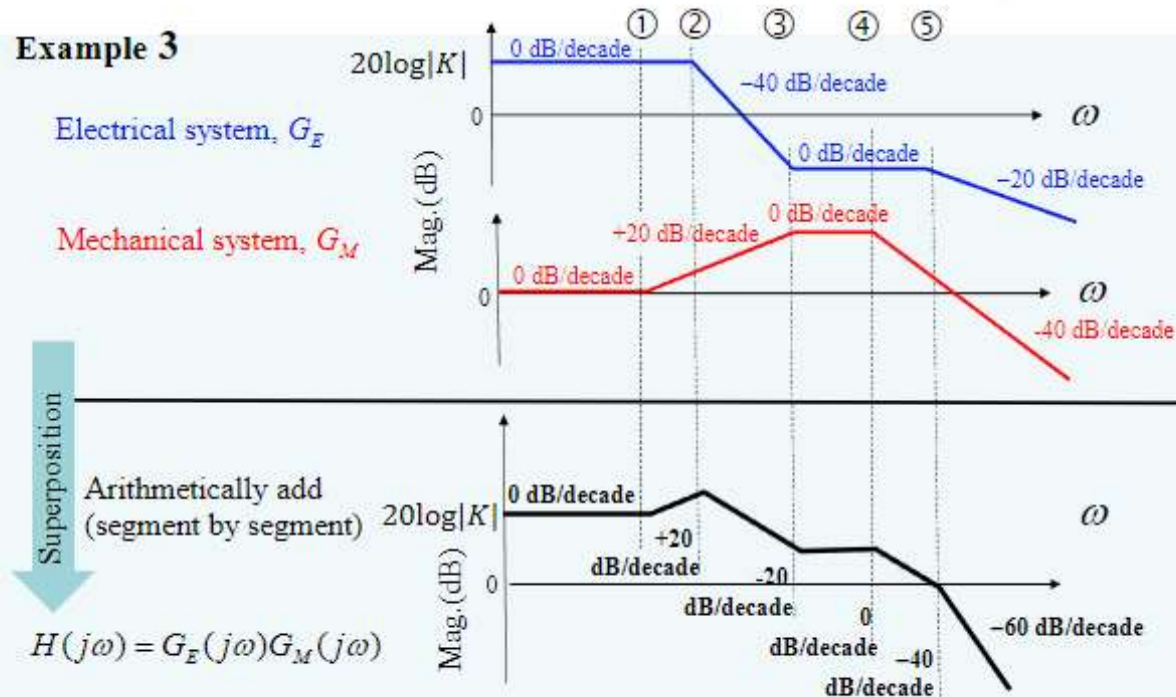
III. Superposition on Bode plots



Example 3

Electrical system, G_E

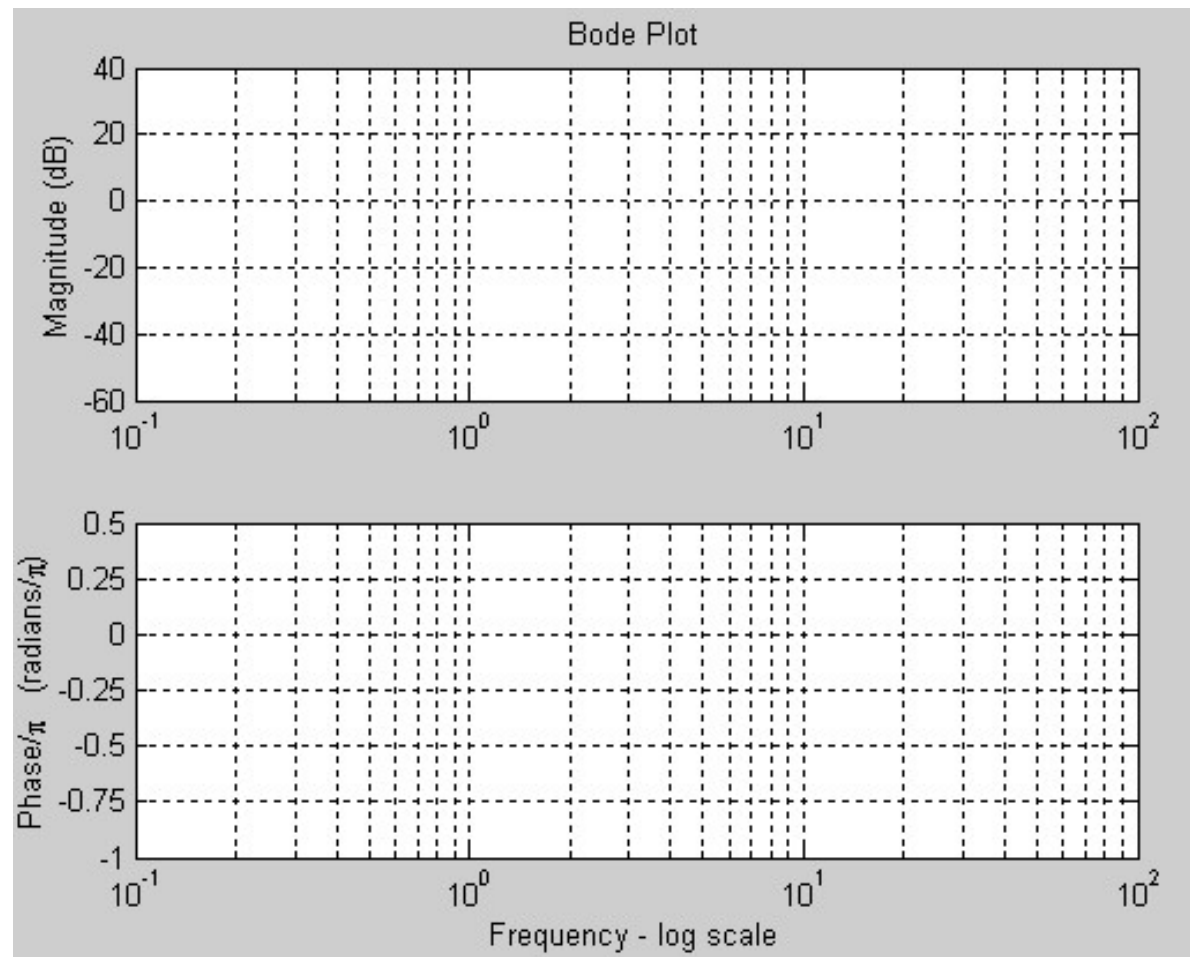
Mechanical system, G_M



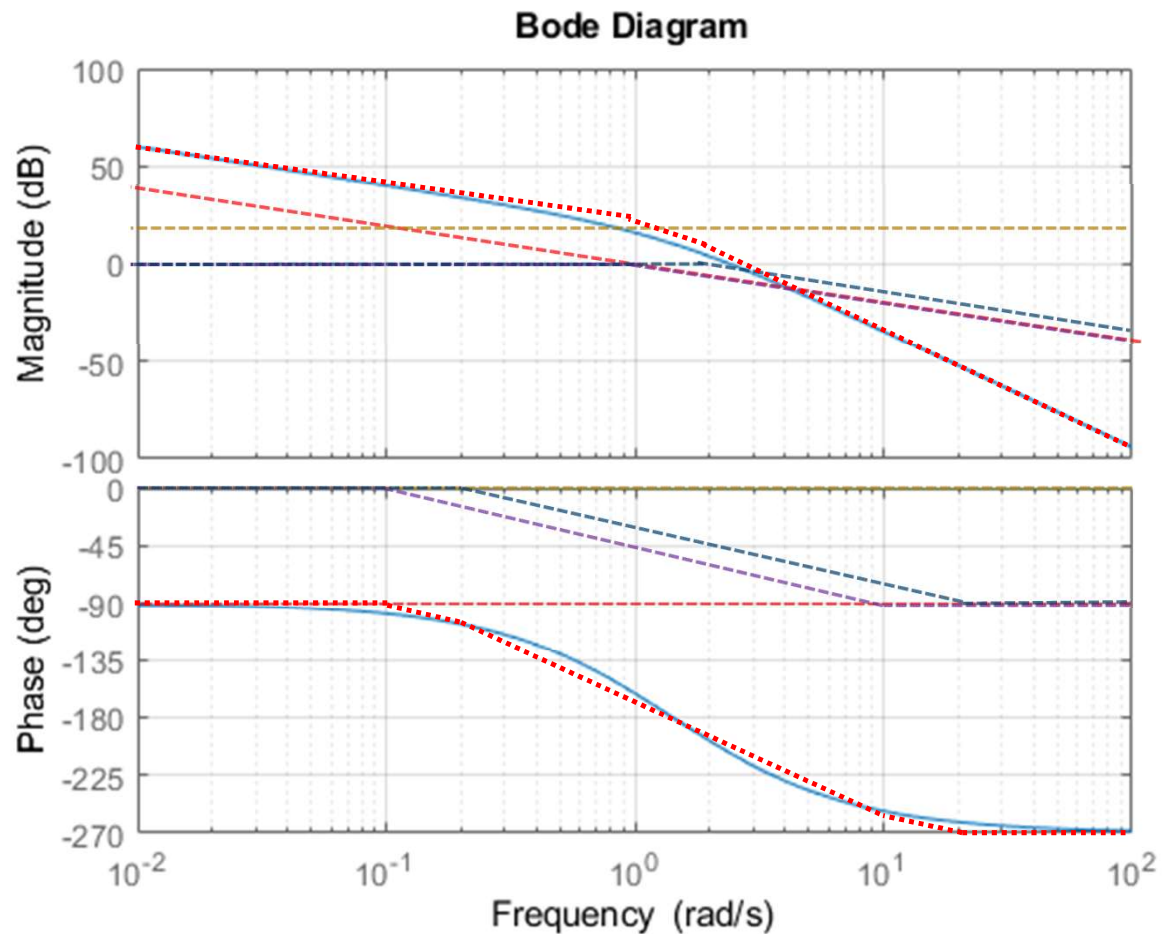
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Bode Diagram - $G(s) = \frac{10}{s(s+1)(0.5s+1)}$



Bode Diagram - $G(s) = \frac{10}{s(s+1)(0.5s+1)}$



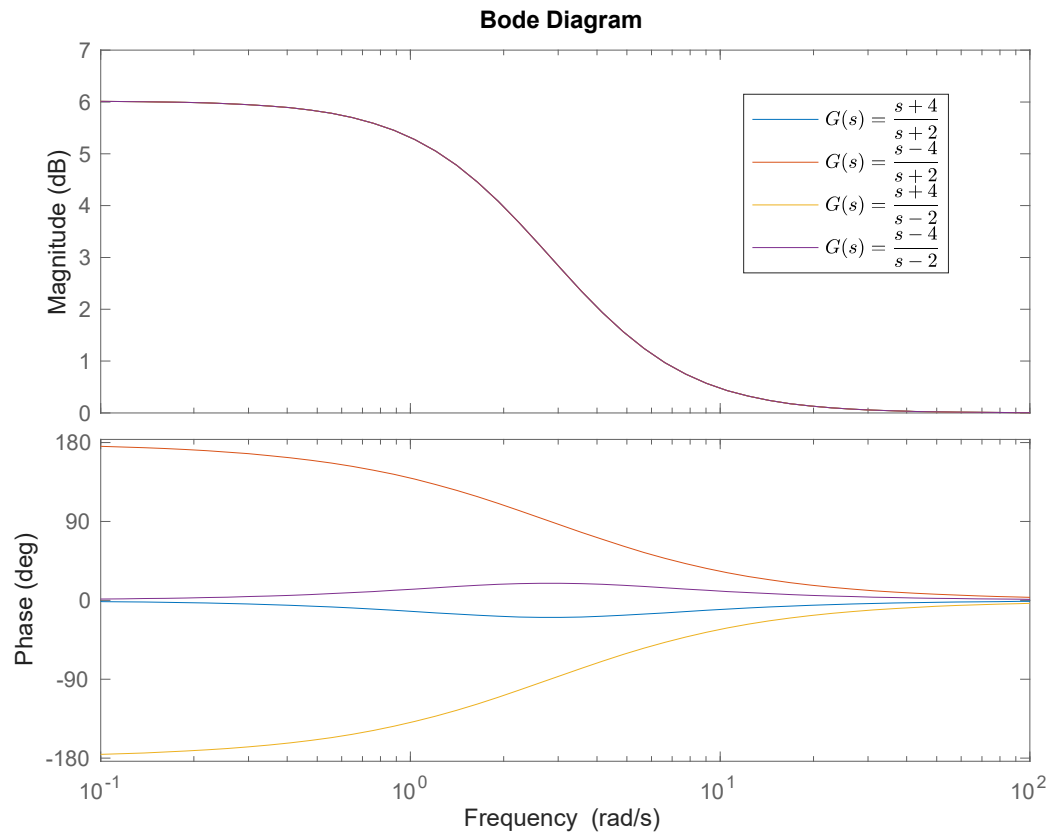
$$K = 10$$

$$G(s) = \frac{1}{s}$$

$$G(s) = \frac{1}{s+1}$$

$$G(s) = \frac{1}{0.5s+1}$$

Bode Diagram – Effect of Switching Sign: ex) $G(s) = \frac{s \pm 4}{s \pm 2}$



1. Sign does not affect the magnitude
2. **But it affects phase**

Lag Compensator

Position $K_{pos} = \lim_{s \rightarrow 0} G(s) = K$

Velocity $K_{vel} = \lim_{s \rightarrow 0} sG(s)$

Acceleration $K_{acc} = \lim_{s \rightarrow 0} s^2 G(s)$

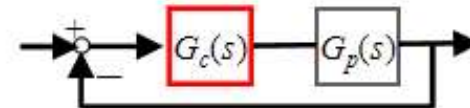
Error constant (Bode Mag. Plot)

$$20 \log |K(j\omega)^{-N}| = 20 \log K_i$$

where $i = pos, vel, \text{ or } acc$

Lag Compensator (Design procedure)

Compensator: $G_c(s) = K \frac{s/a + 1}{s/b + 1} = \left(K \frac{b}{a} \right) \frac{s + a}{s + b}$



To **meet an error constant spec** but keep **phase margin and bandwidth constant**:

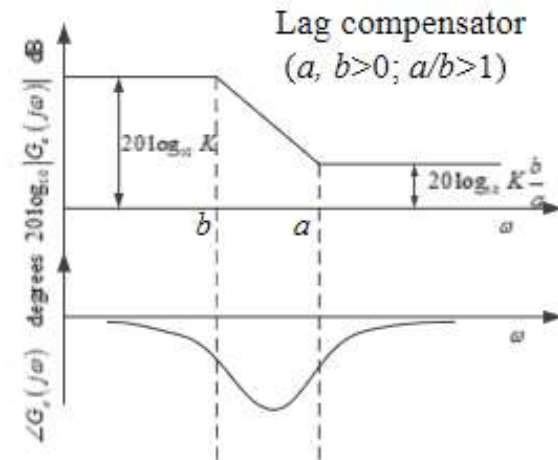
Step 1: Determine the required increase in Bode gain and set the compensator gain K to this value.

Step 2: Let the frequency ratio equal to K .
 $\frac{a}{b} = K$; $K \frac{b}{a} = 1$ at high frequency.

Step 3: Choose the “zero” frequency “ a ” at least a decade below the cross-over frequency ω_c to avoid phase lag contribution at ω_c ($a \leq 0.1\omega_c$)

Step 4: The frequencies “ b ” can be found from Steps 2 and 3: $b = \underbrace{(b/a)}_{1/K} a = a/K$

$$G_c(s) = K \frac{10s/\omega_c + 1}{10Ks/\omega_c + 1}$$



Lead Compensator

Lead Compensator (Design procedure)

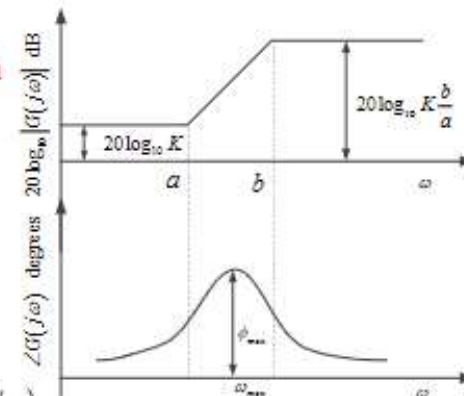
To meet bandwidth and phase margin specs with an unspecified change in gain (error constant).

Lead compensator ($a, b > 0$; $a/b < 1$)

Maximum phase lead $\phi_{\max}$, $\sin \phi_{\max} = \frac{b/a - 1}{b/a + 1}$

Frequency for maximum lead, $\omega_{\max}^2 = ab$

Gain of $\frac{s/a + 1}{s/b + 1}$ at $\omega = \omega_{\max}$, $20 \log_{10} |G(j\omega_{\max})| = 10 \log_{10} (b/a)$



Step 1: Draw the open-loop Bode plots of the uncompensated system.

Step 2: Determine the phase lead required at the specified cross-over frequency to meet the phase margin spec.

Step 3: Let this required lead angle be $\phi_{\max}$ and solve (2) for b/a .

Step 4: Let this maximum lead occur at ω_c so that (3) becomes $\omega_{\max}^2 = \omega_c^2 = ab$. Use this equation and the result of Step 3 to find "a" and "b":

Step 5: Graphically add the frequency response of the lead compensator to the Bode magnitude portion of the uncompensated system.

Step 6: Adjust the gain K of the compensator until the magnitude of the compensated system goes through 0 dB at $\omega = \omega_c$.

$$G_c(s) = K \frac{s/a + 1}{s/b + 1} = \left(K \frac{b}{a} \right) \frac{s + a}{s + b}$$

$$|G(j\omega)| = |G(s)| = 1$$

Sketching Nyquist Plot

Prerequisites for Sketching Nyquist plot

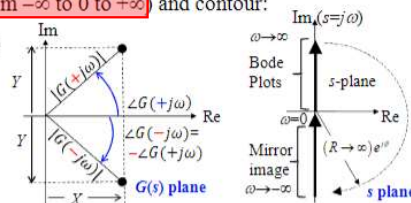
I. Frequency range (ω from $-\infty$ to 0 to $+\infty$) and contour:

$$G(\pm j\omega) = |G(\pm j\omega)| \angle G(\pm j\omega)$$

Complex conjugate

$$|G(\pm j\omega)| = |G(j\omega)|$$

$$\angle G(-j\omega) = -\angle G(j\omega)$$

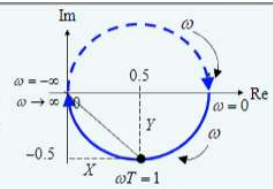


Example: 1st order System

$$G(\pm j\omega) = \frac{1}{\pm j\omega T + 1} = \frac{1}{\sqrt{T^2\omega^2 + 1}} \angle (\mp \tan^{-1} \omega T)$$

$$G(j\omega) = \frac{1}{1 \pm j\omega T} = \frac{1}{1 + T^2\omega^2} + j \frac{\mp \omega T}{1 + T^2\omega^2}$$

$$\left(X - \frac{1}{2}\right)^2 + Y^2 = \left(\frac{1}{2}\right)^2$$



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Prerequisites for Sketching Nyquist plot (cont.)

II. Behavior at $\omega \rightarrow 0$ (System Type)

$$G(j\omega) = \frac{K}{(j\omega)^N} \left[\frac{(j\omega\tau_1 + 1)(j\omega\tau_2 + 1) \cdots (j\omega\tau_m + 1)}{(j\omega T_1 + 1)(j\omega T_2 + 1) \cdots (j\omega T_n + 1)} \right]$$

System type N , number of free integrators

$$G(j\omega) \big|_{\omega \rightarrow 0} = K (j\omega)^{-N} \quad \angle G(j\omega) \big|_{\omega \rightarrow 0} = -90^\circ N$$

$$G(j\omega) = \frac{1}{(j\omega T_1 + 1)(j\omega T_2 + 1)(j\omega T_3 + 1)} \quad \omega \rightarrow \infty \quad \angle G(j\omega) = 0^\circ$$

Type 0 ($N=0$, no free integrator)

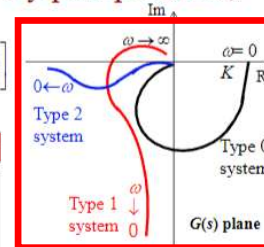
$$|G(j\omega)| = 1, \angle G(j\omega) = 0^\circ$$

Type 1 ($N=1$, Integrator)

$$|G(j\omega)| \rightarrow \infty, \angle G(j\omega) = -90^\circ$$

$$G(j\omega) = \frac{1}{j\omega} \quad \omega \rightarrow \infty \quad \angle G(j\omega) = -90^\circ$$

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Type 2 ($N=2$, double integrators)

$$G(j\omega) = \frac{1}{(j\omega)^2} \quad \omega \rightarrow \infty \quad \angle G(j\omega) = -180^\circ$$

$$|G(j\omega)| \rightarrow \infty, \angle G(j\omega) = -180^\circ$$

$$G(j\omega) = \frac{1}{(j\omega)^2} \quad \omega \rightarrow \infty \quad \angle G(j\omega) = -180^\circ$$

$$|G(j\omega)| \rightarrow \infty, \angle G(j\omega) = -180^\circ$$

$$G(j\omega) = \frac{1}{(j\omega)^2} \quad \omega \rightarrow \infty \quad \angle G(j\omega) = -180^\circ$$

$$|G(j\omega)| \rightarrow \infty, \angle G(j\omega) = -180^\circ$$

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$$G(j\omega) = \frac{1}{(j\omega)^2} \quad \omega \rightarrow \infty \quad \angle G(j\omega) = -180^\circ$$

Prerequisites for Sketching Nyquist plot (cont.)

III. Behavior at $\omega \rightarrow \infty$ (for system where $n \geq m$)

$$G(j\omega) = \frac{K(j\omega\tau_1 + 1)(j\omega\tau_2 + 1) \cdots (j\omega\tau_m + 1)}{(j\omega)^N(j\omega T_1 + 1)(j\omega T_2 + 1) \cdots (j\omega T_n + 1)} \leftarrow O(m)$$

$$G(j\omega) \big|_{\omega \rightarrow \infty} = K (j\omega)^{-(n-m)} \quad \angle G(j\omega) \big|_{\omega \rightarrow \infty} = -90^\circ (n-m)$$

$$G(j\omega) = \frac{j\omega T + 1}{j\omega a T + 1} \leftarrow m = 1 \quad |G(j\omega)| \rightarrow 1/a \quad \angle G(j\omega) = 0^\circ$$

$$n-m=0$$

$$G(j\omega) = \frac{1}{j\omega} \leftarrow m = 0 \quad n-m=1$$

$$|G(j\omega)| \rightarrow 0 \quad \angle G(j\omega) = -90^\circ$$

$$G(j\omega) = \frac{j\omega T_1 + 1}{j\omega(j\omega T_2 + 1)(j\omega T_3 + 1)} \leftarrow m = 1 \quad n-m=2$$

$$|G(j\omega)| = 0; \angle G(j\omega) = -180^\circ$$

$$G(j\omega) = \frac{1}{(j\omega T_1 + 1)(j\omega T_2 + 1)(j\omega T_3 + 1)} \leftarrow m = 0 \quad n-m=3$$

$$|G(j\omega)| = 0; \angle G(j\omega) = -270^\circ$$

$$G(j\omega) = \frac{1}{(j\omega T_1 + 1)(j\omega T_2 + 1)(j\omega T_3 + 1)} \leftarrow m = 0 \quad n-m=3$$

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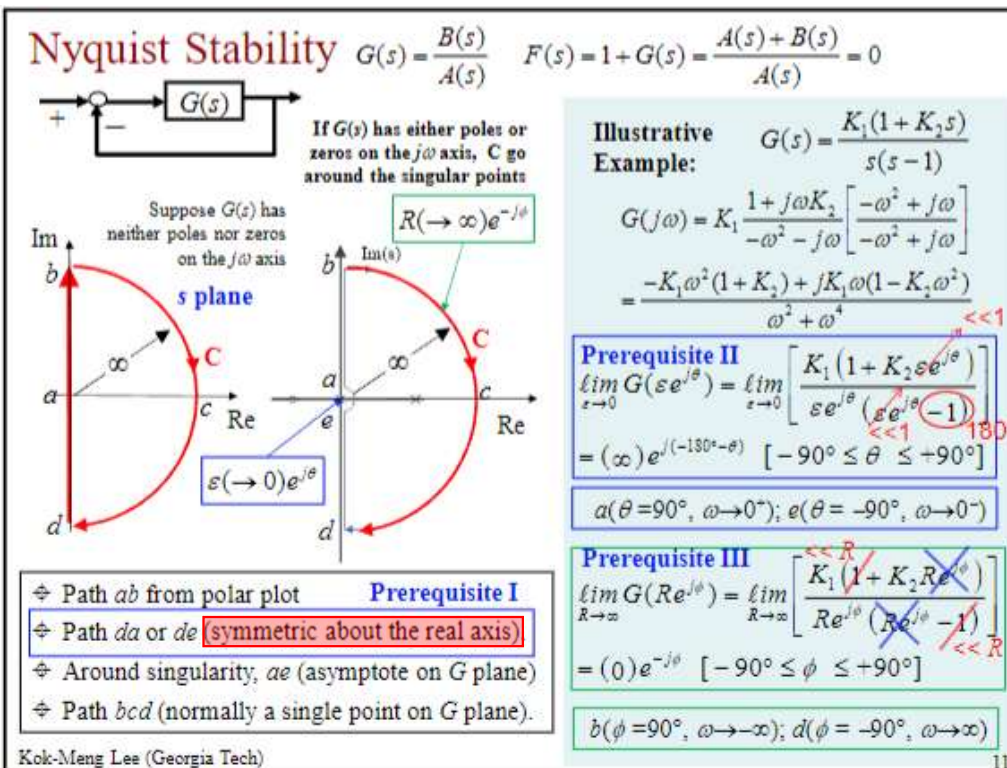
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Kok-Meng Lee (Georgia Tech)

Sketching Nyquist Plot



Phase margin and Gain margin (Minimum phase system)

Phase margin γ

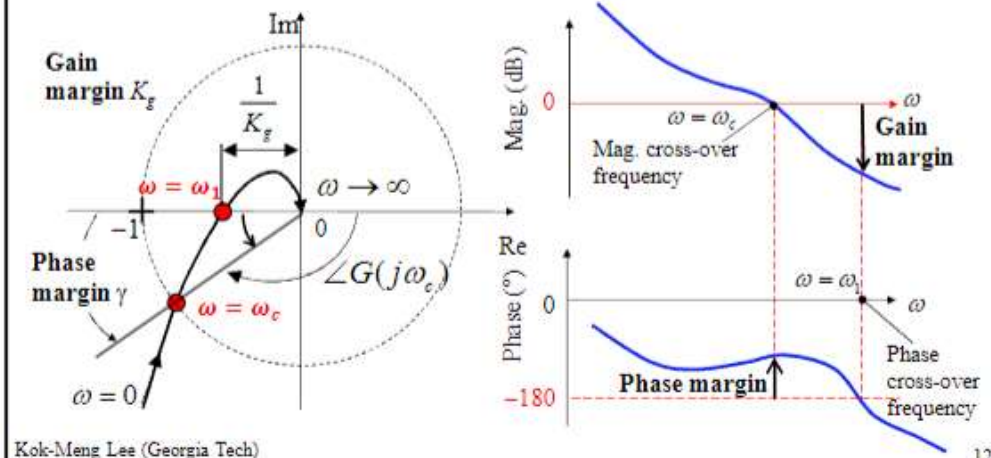
$$\gamma = 180^\circ + \angle G(j\omega_c)$$

$$|G(j\omega_c)| = 1 \text{ or } 0 \text{ dB}$$

Gain margin K_g

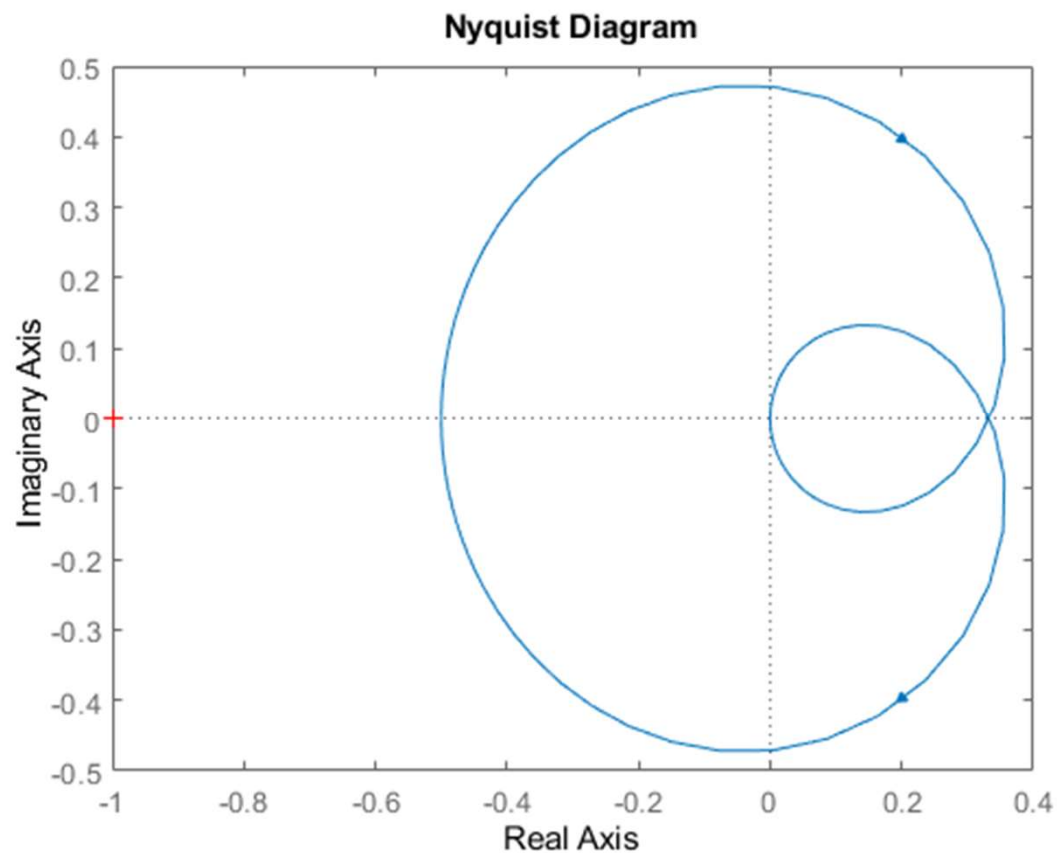
$$K_g = \frac{1}{|G(j\omega_1)|} \quad K_g = -20 \log |G(j\omega_1)|$$

$$\angle G(j\omega_1) = -180^\circ$$



Example - $G(s) = \frac{s-1}{(s+2)(s+1)} = \frac{4\omega^2-2}{\omega^4+5\omega^2+4} - \frac{j\omega(\omega-5)}{\omega^4+5\omega^2+4}$

Example - $G(s) = \frac{s-1}{(s+2)(s+1)} = \frac{4\omega^2-2}{\omega^4+5\omega^2+4} - \frac{j\omega(\omega-5)}{\omega^4+5\omega^2+4}$



Nyquist Criteria - $G(s) = \frac{s-1}{(s+2)(s+1)}$

Nyquist Stability theorem:

$$N = Z - P \Rightarrow Z = N + P$$

P = # of unstable open-loop poles
[or poles of $F(s)$]

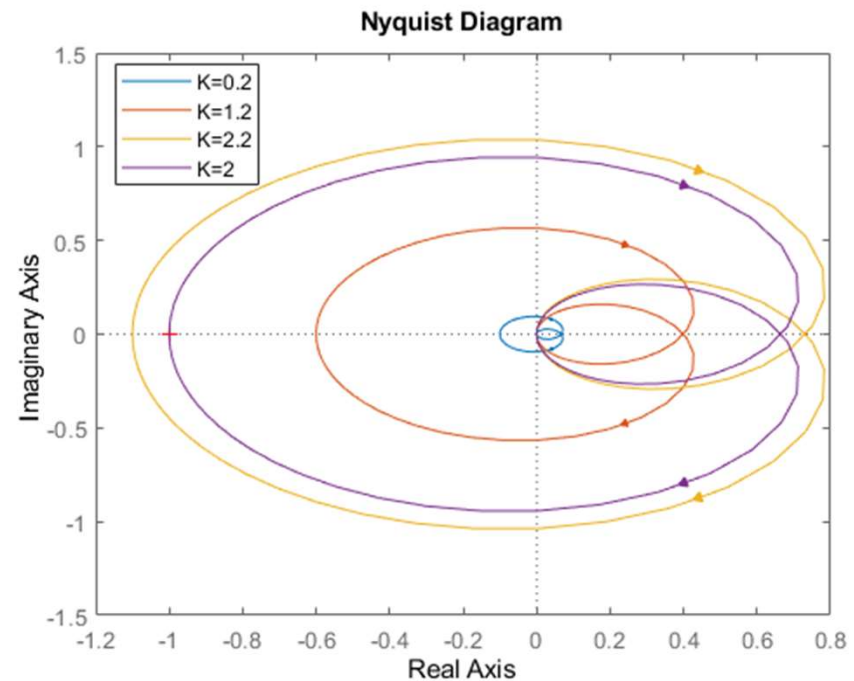
Z = # of unstable closed-loop poles
[or zeros of $F(s)$]

N = # of clockwise rotations of $F(s)$
around the origin of the $F(s)$ plane

= # of clockwise rotations of $G(s)$
around $(-1, j0)$.

1) This is normally used to find Z
based on P and N .

2) For stability, $Z=0$.



Verify whether Z changes from 0 to another value.

State Space

	System matrix, $n \times n$	Input matrix, $n \times r$	
	↓	↓	
State equation:	$\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$		$\mathbf{x} = n \times 1$ State vector
			$\mathbf{u} = r \times 1$ Input vector
Output equation:	$\mathbf{y} = \mathbf{Cx} + \mathbf{Du}$		$\mathbf{y} = m \times 1$ Output vector
	↑	↑	
	Output matrix, $m \times n$	Direct transmission matrix, $m \times r$	

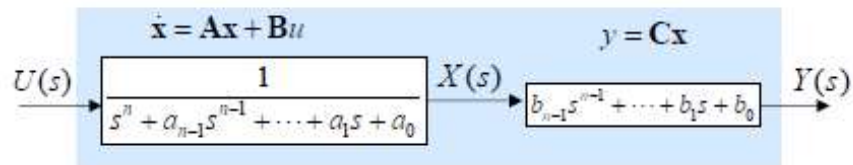
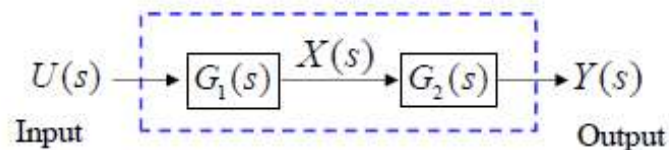
Transfer function to State Space

From transfer function (SISO) to state-space equation

$$G(s) = \frac{b_{n-1}s^{n-1} + \dots + b_1s + b_0}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0} = \frac{Y(s)}{U(s)} = \underbrace{\frac{X(s)}{U(s)}}_{G_1(s)} \underbrace{\frac{Y(s)}{X(s)}}_{G_2(s)}$$

$$\text{Let } \frac{X(s)}{U(s)} = \frac{1}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

$$\frac{Y(s)}{X(s)} = b_{n-1}s^{n-1} + \dots + b_1s + b_0$$



State-space representation (SISO)

$$\frac{X(s)}{U(s)} = \frac{1}{s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0}$$

$$\frac{d^n x}{dt^n} + a_{n-1} \frac{d^{n-1}x}{dt^{n-1}} + \dots + a_1 \frac{dx}{dt} + a_0 x = u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u$$

State vector

Let

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x \\ \frac{dx}{dt} = \dot{x}_1 \\ \frac{d^2x}{dt^2} = \dot{x}_2 \\ \vdots \\ \frac{d^{n-1}x}{dt^{n-1}} = \dot{x}_{n-1} \end{bmatrix}$$

State equation: $\dot{\mathbf{x}} = \mathbf{Ax} + \mathbf{Bu}$

With zero initial condition

$$\frac{dx(t)}{dt} \leftrightarrow sX_I(s)$$

$$\frac{d^2x(t)}{dt^2} \leftrightarrow s^2X_I(s)$$

$$\frac{d^nx(t)}{dt^n} \leftrightarrow s^nX_I(s)$$

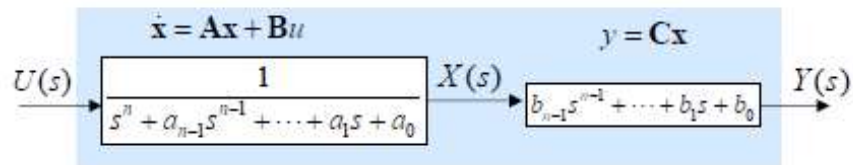
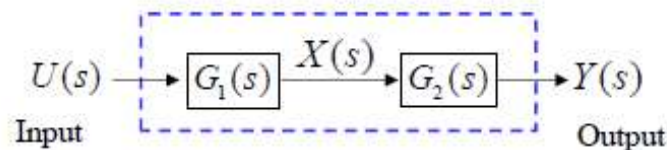
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State Space to Transfer Function

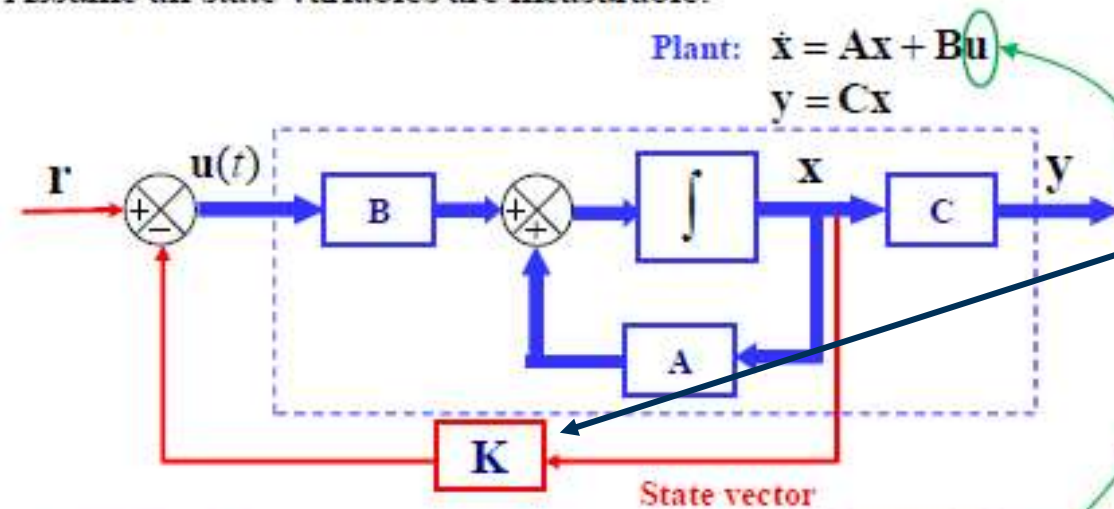
$$\begin{array}{l} \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u \\ y = \mathbf{C}\mathbf{x} \end{array} \quad \longrightarrow \quad G(s) = \frac{Y(s)}{U(s)} = \underset{\substack{\uparrow \\ 1 \times n}}{\mathbf{C}} \underset{\substack{\uparrow \\ n \times n}}{(s\mathbf{I} - \mathbf{A})}^{-1} \underset{\substack{\uparrow \\ n \times 1}}{\mathbf{B}}$$

Characteristic equation: $\det [s\mathbf{I} - \mathbf{A}] = 0$

Full State Feedback Control

Full State feedback control (pole placement)

Assume all state variables are measurable.



Gain may not be constant

□ Closed-loop system

Control law: $u(t) = -Kx(t) + r$

$$\dot{x} = (A - BK)x + Br$$

Closed loop characteristic equation: $|sI - (A - BK)| = 0$

Desired closed-loop poles can be placed using pole placement method.